

1 Very little of the light energy falling on a field of plants is absorbed by the plants and transferred to higher trophic levels.

(a) Suggest **two** reasons why very little of the light energy is absorbed by plants.

(2)

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(b) In an American study, the light energy falling on one square metre of a field in one year was recorded as 1 700 000 kilocalories.

(i) Only one fifteenth of this light energy was converted into plant biomass.

Calculate the number of kilocalories of light energy converted into plant biomass.

Give your answer in standard form.

(1)

Answer kilocalories

- (ii) Only 6 kilocalories became animal biomass.

Calculate the number of kilojoules (kJ) of energy that became biomass.

1 kilocalorie = 4.2 kJ.

Give your answer to **two** significant figures.

(1)

Answer kJ

- (iii) In another field 154 000 kilocalories of energy was converted into plant biomass.

Only 25 kilocalories of this energy was passed onto the highest trophic level in a food chain.

Calculate the percentage of energy that was passed onto the highest trophic level.

(1)

Answer %

(Total for Question 1 = 5 marks)